



MASTERMIND --- REVISED

Overview

Mastermind is a code-breaking game in which players attempt to discover a secret color code.

The game supports one to four players and provides multiple configuration options that affect gameplay.

The objective of the game is to correctly determine the secret code before the game ends.

Game Configuration

Before a game begins, the following options are specified.

Number of Players

Supported values:

- 1 player
- 2 players
- 3 players
- 4 players

Codemaker Mode

The game supports two codemaker modes.

System Codemaker Mode

The system acts as the Codemaker.

For each game, the system generates a secret code that remains hidden from all players.

Players attempt to discover the secret code.

This is the default mode.

Player Codemaker Mode

Players alternate between the roles of Codemaker and Codebreaker.

For each round:

- One player serves as the Codemaker.
- One player serves as the Codebreaker.
- The Codemaker creates a secret code using the configured game options.
- The secret code remains hidden from the Codebreaker.
- The Codebreaker attempts to discover the code.

At the conclusion of a round, the Codemaker role rotates to another player.

A complete game consists of enough rounds to allow each player to serve as Codemaker exactly once.

The game ends after all players have completed a turn as Codemaker.

Code Length

The secret code may contain:

- 4 pegs (default)
- 5 pegs
- 6 pegs

Color Repetition

Two repetition modes are supported.

Repetition Allowed

A color may appear multiple times within the secret code.

Example:

Red, Blue, Red, Yellow

This is the default mode.

Unique Colors Only

Each color may appear at most once within the secret code.

Example:

Red, Blue, Yellow, Green

The following code would not be valid:

Red, Blue, Red, Yellow

AVAILABLE COLORS

Secret codes and guesses may use the following colors:

- Red
- Orange
- Yellow
- Green
- Blue
- Purple
- Brown
- Lavender

STARTING A GAME

When a new game begins:

1. The game configuration is specified.
2. A secret code is established according to the selected codemaker mode.
3. The secret code remains hidden from the Codebreaker.
4. Play begins.

TAKING A TURN

During a turn:

1. The Codebreaker submits a guess.
2. The guess must contain the configured number of pegs.
3. The guess must use only valid colors.
4. The guess must satisfy the selected color repetition rules.
5. The guess is evaluated against the secret code.
6. Feedback is provided.

FEEDBACK

Feedback consists of two types of information.

Correct Color in Correct Position

A marker is awarded for each guessed color that matches both the color and position of a code peg.

Correct Color in Incorrect Position

A marker is awarded for each guessed color that exists in the secret code but appears in a different position.

The order of feedback markers does not indicate which specific pegs were matched.

WINNING A ROUND

A Codebreaker wins a round by correctly identifying every color and position in the secret code.

When a winning guess is submitted:

1. The round immediately ends.
2. The secret code is revealed.
3. The winner is identified.

ENDING A ROUND

A round ends when:

- The Codebreaker successfully discovers the secret code, or
- The maximum number of guesses has been reached.

If the secret code is not discovered, the secret code is revealed at the end of the round.

ENDING A GAME

System Codemaker Mode

The game ends when:

- A player discovers the secret code, or

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- The maximum number of guesses has been reached.

Player Codemaker Mode

The game ends after every player has completed one round as Codemaker.

The game then determines the final results based on player performance across all rounds.